

Copula Entropy

Theory and Applications

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- Lecture 7: Generalizations
- **Lecture 8: Real applications and softwares**

Lecture 8: Real applications and softwares

1 Real applications

- Astronomy
- Physics
- Chemistry
- Geology
- Biology
- Medicine
- Social sciences
- Engineering

2 Softwares

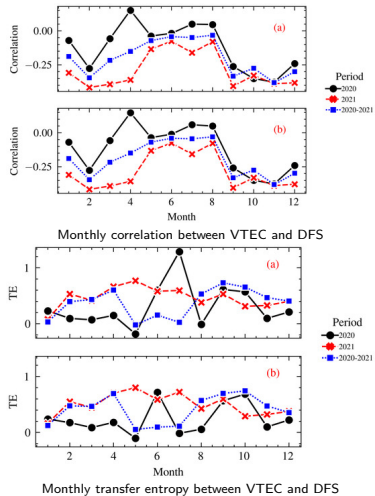
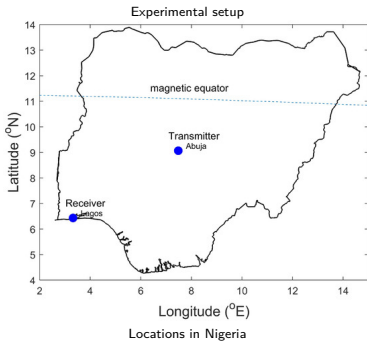
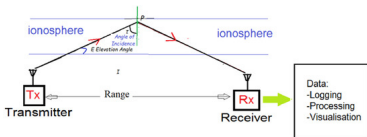
- Official
- Third-party

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- 赤道电离层特性随季节变化分析
 - 基于 TE 分析赤道地区高频微波信号传输中 TEC 和 DFS 之间的非线性关系³

³ Aderonke Akerele et al. "Complexity and Nonlinear Dependence of Ionospheric Electron Content and Doppler Frequency Shifts in Propagating HF Radio Signals within Equatorial Regions". In: *Atmosphere* 15.6 (2024), p. 654. DOI: 10.3390/atmos15060654. URL: <https://www.mdpi.com/2073-4433/15/6/654>.

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热学

- 微热管设计和制造
 - 提出了一种基于 CE 和神经网络模型相结合的微热管设计和制造参数预测方法⁶
- 循环流化床锅炉工艺参数优化
 - 结合 CE、加权融合树模型和多目标树结构估计器的循环流化床锅炉工艺参数优化方法⁷

⁶ 李勇 et al. “基于一维卷积残差网络的微热管几何结构及制造工艺参数预测”. In: 中国科学: 技术科学 55.2 (2025), pp. 281–294, 张靖昊. “基于机器学习的微热管结构及工艺参数设计系统开发”. 硕士学位论文. 华南理工大学, 2024.

⁷ Yunpeng Ma et al. “Improve thermal efficiency and reduce NOx emission of circulating fluidized bed boiler based on multi-objective optimization framework”. In: *Thermal Science and Engineering Progress* (2025), p. 103753. ISSN: 2451-9049. DOI: <https://doi.org/10.1016/j.tsep.2025.103753>. URL: <https://www.sciencedirect.com/science/article/pii/S2451904925005438>.

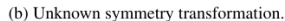
- 变构效应研究
 - 变构效应配位点和激活点热力学耦合模型⁸

⁸Michel A. Cuendet, Harel Weinstein, and Michael V. LeVine. "The Allosteric Landscape: Quantifying Thermodynamic Couplings in Biomolecular Systems". In: *Journal of Chemical Theory and Computation* 12.12 (2016), pp. 5758–5767. ISSN: 1549-9618. DOI: 10.1021/acs.jctc.6b00841. URL: <https://doi.org/10.1021/acs.jctc.6b00841>.

- 分子设计
 - 设计具有特定属性的分子结构⁹

⁹Mario Wieser. “Learning Invariant Representations for Deep Latent Variable Models”. PhD thesis. University of Basel, 2020, Mario Wieser et al. “Inverse Learning of Symmetries”. In: *Advances in Neural Information Processing Systems*. Vol. 33, 2020, pp. 18004–18015.

✓ ✓ ✓

[illegible]

¹¹ Song Zhang et al. "Correlation between plastic deformation mechanism and texture evolution of a near β -Ti alloy deformed in β region". In: *Intermetallics* 170 (2024), p. 108333. ISSN: 0966-9795. DOI: <https://doi.org/10.1016/j.intermet.2024.108333>. URL: <https://www.sciencedirect.com/science/article/pii/S0966979524001523>.

- 大气污染传播路径分析
 - 兰州市大气污染传播路径预测¹³
- 火电厂排放污染物管控
 - 火电厂氮氧化物排放浓度预测¹⁴
- 氨气污染
 - 长三角地区氨气浓度长期变化及其驱动因素¹⁵
- SO_2 排放浓度预测
 - 提出了一种 SO_2 提前预测的软测量技术¹⁶

¹³ 吴京鹏. “基于图嵌入表示的节点无特征网络链路预测研究”. 硕士学位论文. 西北师范大学, 2022.

¹⁴ 金秀章, 乔鹏, and 史德金. “基于 VMD-Bayes-Lasso 算法带误差补偿的火电厂 NO_x 浓度软测量”. In: 华北电力大学学报 (自然科学版) 52 235 (2023), pp. 117–124+142.

¹⁵ Jingkai Xue et al. "Long-term spatiotemporal variations of ammonia in the Yangtze River Delta region of China and its driving factors". In: *Journal of Environmental Sciences* 150 (2025), pp. 202–217. ISSN: 1001-0742. DOI: <https://doi.org/10.1016/j.jes.2024.02.021>. URL: <https://www.sciencedirect.com/science/article/pii/S1001074224001153>.

¹⁶ 乔鹏. “基于 CatBoost-Bayes 带误差补偿的脱硫出口 SO_2 浓度软测量研究”. 硕士学位论文. 华北电力大学, 2024.

地质学

- 岩相分类
 - 利用 CE 构建岩相分类模型¹⁷

¹⁷Jian Ma. "Facies Classification with Copula Entropy". In: *arXiv preprint arXiv:2501.14351* (2025).

- 土地干旱度分类
 - 考虑蒸散过程的土地干旱度分类¹⁸
- 北欧山地陆气耦合分析
 - 利用 CE 分析了芬瑟两个观测站的气象观测变量之间 CE 和解耦度量之间的关系¹⁹

¹⁹Laura Mack, Marvin Kähnert, and Norbert Pirk. “Probabilistic modelling of atmosphere-surface coupling with a copula Bayesian network”. In: *Environmental Modelling & Software* (2026). arXiv: 2509.11975 [physics.ao-ph]. URL: <https://arxiv.org/abs/2509.11975>.

CASE 4: Atmosphere-surface coupling by Oslo University

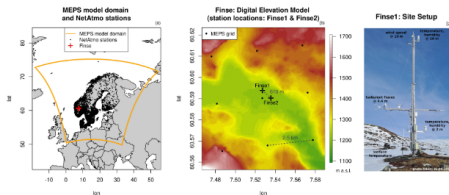


Figure 1: Used data sources: (a) Model domain of the NWP model MEPS and locations of the NetAtmo stations. (b) Surrounding of Finse and station locations. (c) Setup of the main site Finse1.

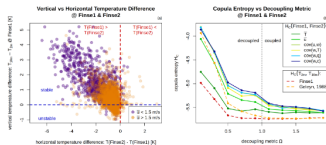


Figure 2: Relation of vertical and horizontal coupling: (a) Horizontal temperature difference between Finse1 and Finse2 versus vertical temperature difference at Finse1 colored based on a wind speed threshold (30 minutes temporal average), (b) decoupling metric at Finse1 versus copula entropy between Finse1 and Finse2 of temperature, wind speed, and the covariances corresponding to the fluxes of momentum $\text{cov}(u,w)$, sensible heat $\text{cov}(w,T)$, latent heat $\text{cov}(q,w)$ and CO_2 $\text{cov}(c,w)$ (30 minutes temporal average). Entropy calculation is based on time series of the same length to avoid a sampling size bias.

水文学 1

- 洪水预报
 - 金沙江流域洪水预报²⁰
 - 天山北麓奎屯河洪水分析²¹
- 河流相关性
 - 长江上游河段（金沙江、岷江、沱江、嘉陵江）相关性²²
 - 长江上游河段复合洪水事件分析²³
- 水沙关系分析
 - 黄河西柳沟河流域径流量和输沙量数据分析²⁴

²⁰ Lu Chen, Vijay P. Singh, and Shenglian Guo. "Measure of Correlation between River Flows Using the Copula-Entropy Method". In: *Journal of Hydrologic Engineering* 18.12 (2013), pp. 1591-1606, Lu Chen et al. "Copula entropy coupled with artificial neural network for rainfall-runoff simulation". In: *Stochastic Environmental Research and Risk Assessment* 28.7 (2014), pp. 1755-1767, 陈佳雷 et al. "一种基于多要素注意力时空图卷积网络的径流预报方法". Pat. CN111715128A, CN111715128A, 2023.

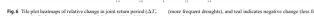
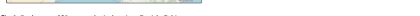
²¹Yingying Han et al. "Non-Stationary Flood Characteristics and Joint Risk Analysis in Inland China with Uncertainty Considerations". In: *Atmosphere* 17.3 (2026). p. 281.

²²Lu Chen and Shenglian Guo. *Copulas and its Application in Hydrology and Water Resources*. Springer Singapore, 2018.

²³Xu Wang and Yong-Ming Shen. "A Framework of Dependence Modeling and Evaluation System for Compound Flood Events". In: *Water Resources Research* 59.8 (2023). e2023WR034718. DOI: <https://doi.org/10.1029/2023WR034718>.

²⁴Longxia Qian et al. "A New Estimation Method for Copula Parameters for Multivariate Hydrological Frequency Analysis With Small Sample Sizes". In: *Water Resources Management* 36.4 (2022), pp. 1141–1157. ISSN: 1573-1650. URL: <https://doi.org/10.1007/s11269-021-03016-w>.

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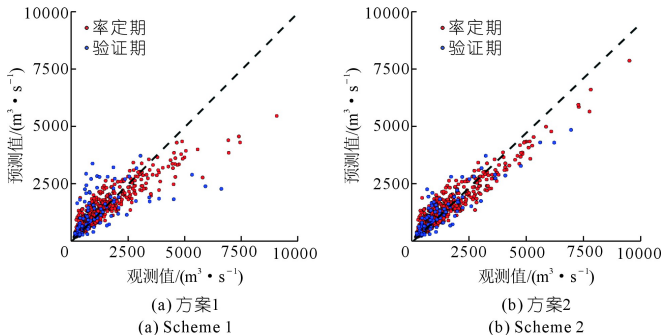


Figure: Comparison between observations and predictions by correlation coefficients and copula entropy on monthly inflow in Danjiangkou reservoir.

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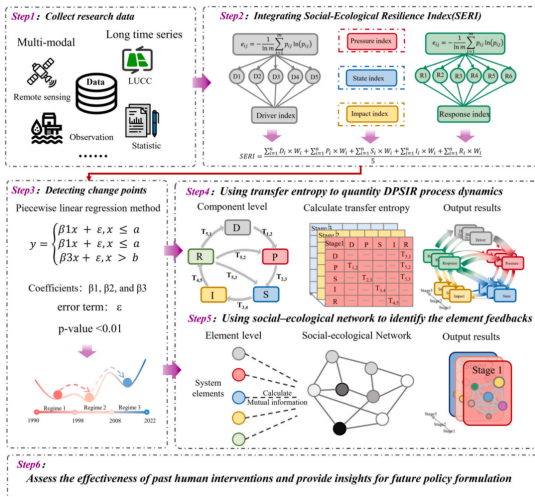
⁴⁵ Francesco Gardino, "La divergenza di Jensen-Shannon nell'algoritmo di clustering dinamico per dati descritti da distribuzioni", <https://arxiv.org/abs/2006.07001>.

- 动物运动轨迹分析
 - Cylcop 算法包⁵⁰
- 土壤生态学
 - 利用微生物-酶分解模型研究了土壤湿度变化对土壤碳氮动态性的长期影响⁵¹
- 社会生态学
 - 利用基于 CE 的 TE 和网络分析等工具研究西南喀斯特地区社会生态系统脆弱性演化⁵²

⁵⁰Florian H. Hodel and John R. Fieberg. "cylcop: An R Package for Circular-Linear Copulae with Angular Symmetry". In: *bioRxiv* (2021), p. 2021.07.14.452253, Florian Hodel. *cylcop: Circular-Linear Copulas with Angular Symmetry for Movement Data*. CRAN. R package version 0.2.0, URL: <https://cran.r-project.org/package=cylcop>. 2022.

⁵¹Wanyu Li et al. "Microbially-Mediated Soil Carbon-Nitrogen Dynamics in Response to Future Soil Moisture Change". In: *Earth's Future* 13.3 (2025), e2024EF005521. DOI: <https://doi.org/10.1029/2024EF005521>.

⁵²Yueying Li et al. "Integrating Transfer Entropy and Network Analysis to Explore Social-Ecological Resilience Evolution: a case study in South China Karst". In: *Journal of Cleaner Production* (2025), p. 145926. ISSN: 0959-6526. DOI: <https://doi.org/10.1016/j.jclepro.2025.145926>. URL: <https://www.sciencedirect.com/science/article/pii/S0959652625012764>.



“...and the other side of the road.”

⁵⁴Soumik Purkayastha and Peter X.K. Song. "Asymmetric predictability in causal discovery: an information theoretic approach". In: *arXiv preprint arXiv:2210.14455* (2022).

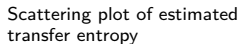
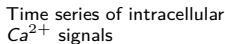
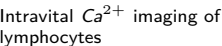
植物学

- 树干液流预测
 - 提出了一种基于历史环境因子预测树干液流的方法⁵⁵

⁵⁵Yane Li et al. "CE-CNN-BiGRU-SA: A novel generalization sap flow prediction model for trees at different ages and species with historical environmental factors". In: *SSRN:5460406* (2025).

⁵⁸王嘉龙, “南极磷虾及其相关种空间分布对未来气候变化的响应研究”, 硕士学位论文, 上海海洋大学, 2024.

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神经科学 1

● 神经病学

- 注意缺陷多动障碍⁶⁰
- 癫痫症⁶¹
- 脑卒中⁶²

⁶⁰Paolo Victor Redondo, Raphaël Huser, and Hernando Ombao. "Measuring information transfer between nodes in a brain network through spectral transfer entropy". In: *The Annals of Applied Statistics* 19.3 (2025), pp. 2386–2411. DOI: [10.1214/25-AOAS2050](https://doi.org/10.1214/25-AOAS2050). URL: <https://doi.org/10.1214/25-AOAS2050>, 汪方毅 et al. "基于静息态 fMRI 区分健康老年人认知水平的 MVPA 方法研究". In: *磁共振成像* 14.6 (2023), pp. 18–25.

⁶¹Zhaohui Li et al. "Estimating global phase synchronization by quantifying multivariate mutual information and detecting network structure". In: *Neural Networks* 183 (2025), p. 106984. ISSN: 0893-6080. DOI: <https://doi.org/10.1016/j.neunet.2024.106984>. URL: <https://www.sciencedirect.com/science/article/pii/S0893608024009134>.

⁶²Wojciech Ciezobka et al. "End-to-End Stroke Imaging Analysis Using Effective Connectivity and Interpretable Artificial Intelligence". In: *IEEE Access* 13 (2025), pp. 10227–10239. DOI: [10.1109/ACCESS.2025.3529179](https://doi.org/10.1109/ACCESS.2025.3529179).

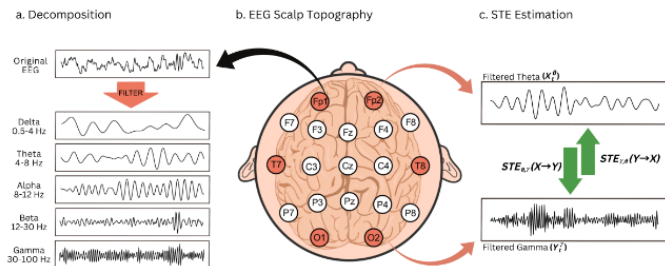


FIG 9. Building blocks of STE. (a) Application of bandpass filter on original EEG series to obtain frequency band-specific filtered series. (b) Standard 10–20 EEG scalp topography where the highlighted channels are included in the analysis. (c) Illustration of STE estimation between two filtered series.

神经科学 2

● 认知神经学

- 分析大脑认知活动的多模态数据⁶³
- 语音信息的编码和解析⁶⁴
- 神经元特异化识别⁶⁵

⁶³Stephanie J. Kayser et al. "Irregular Speech Rate Dissociates Auditory Cortical Entrainment, Evoked Responses, and Frontal Alpha". In: *The Journal of Neuroscience* 35.44 (2015), pp. 14691–14701, Robin A. A. Ince et al. "The Deceptively Simple N170 Reflects Network Information Processing Mechanisms Involving Visual Feature Coding and Transfer Across Hemispheres". In: *Cerebral Cortex* 26.11 (2016), pp. 4123–4135, Robin A.A. Ince et al. "A statistical framework for neuroimaging data analysis based on mutual information estimated via a gaussian copula". In: *Human Brain Mapping* 38.3 (2017), pp. 1541–1573, Etienne Combrisson et al. "Group-level inference of information-based measures for the analyses of cognitive brain networks from neurophysiological data". In: *NeuroImage* 258 (2022), p. 119347. ISSN: 1053-8119. DOI: <https://doi.org/10.1016/j.neuroimage.2022.119347>. URL: <https://www.sciencedirect.com/science/article/pii/S1053811922004669>.

⁶⁴Pieter De Clercq et al. "Beyond linear neural envelope tracking: a mutual information approach". In: *Journal of Neural Engineering* 20.2 (2023), p. 026007. DOI: 10.1088/1741-2552/acbe1d. URL: <https://dx.doi.org/10.1088/1741-2552/acbe1d>.

⁶⁵NA Pospelov et al. "Searching for cognitive specializations of neurons using mutual information framework". In: *Genes & Cells* 18.4 (2023), pp. 878–881.

- 运动神经学
 - 分析运动的肌肉组合协同策略⁶⁶
- 计算神经学
 - 神经元可塑性建模⁶⁷
 - 神经网络信息传输关系分析⁶⁸

66 吴亚婷 et al. “多尺度肌间耦合网络分析”. In: 生物医学工程学杂志 38.4 (2021), pp. 742–752, Yating Wu et al. “R-Vine Copula Mutual Information for Intermuscular Coupling Analysis”. In: *Proceedings of the 11th International Conference on Computer Engineering and Networks*. 2022, pp. 526–534, David O’ Reilly and Ioannis Delis. “A network information theoretic framework to characterise muscle synergies in space and time”. In: *Journal of Neural Engineering* 19.1 (2022), p. 016031. doi: 10.1088/1741-2552/ac5150. url: <https://doi.org/10.1088/1741-2552/ac5150>, Shaojun Zhu et al. “Intermuscular coupling network analysis of upper limbs based on R-vine copula transfer entropy”. In: *Mathematical Biosciences and Engineering* 19.9 (2022), pp. 9437–9456, 金国美 et al. “基于小波包-Copula 互信息的肌间耦合特性”. In: 传感技术学报 35.10 (2022), pp. 1348–1353.

⁶⁷ Johannes Leugering and Gordon Pipa. "A Unifying Framework of Synaptic and Intrinsic Plasticity in Neural Populations.". In: *Neural Computation* 30.4 (2018), pp. 945–986.

⁶⁸ Ari Pakman et al. "Estimating the Unique Information of Continuous Variables in Recurrent Networks". In: *Advances in Neural Information Processing Systems* 34 (2021), pp. 20295–20307.

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医学 2

● 临床医学 2

- 白内障术后角膜水肿风险预测⁷⁸
- 主动脉瓣置换手术射血分数分析⁷⁹
- 脑肿瘤医学影像组学诊断模型构建⁸⁰
- 基于脉搏波的高血压和糖尿病健康状态分类⁸¹

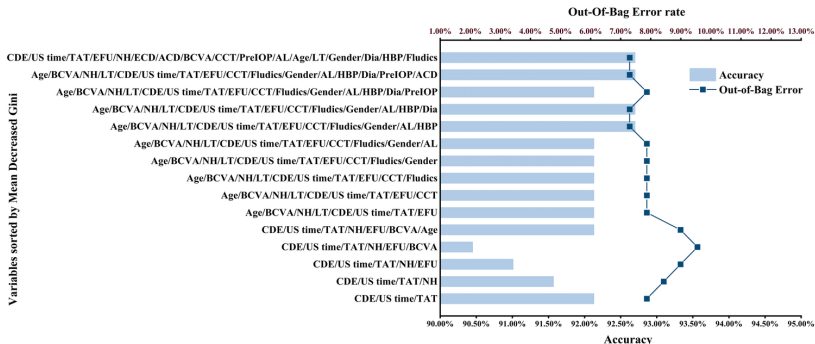
⁷⁸Yu Luo et al. "Research on Establishing Corneal Edema after Phacoemulsification Prediction Model Based on Variable Selection with Copula Entropy". In: *Journal of Clinical Medicine* 12.4 (2023), p. 1290. ISSN: 2077-0383. DOI: 10.3390/jcm12041290. URL: <https://www.mdpi.com/2077-0383/12/4/1290>.

⁷⁹S.M. Sunoj and N. Unnikrishnan Nair. "Survival Copula Entropy and Dependence in Bivariate Distributions". In: *REVSTAT-Statistical Journal* 23.1 (2025), pp. 101–115. URL: <https://revstat.ine.pt/index.php/REVSTAT/article/view/560>.

⁸⁰潘红宇. "基于影像组学与深度学习的脑肿瘤图像分类研究". 硕士学位论文. 西南大学, 2023.

⁸¹汤宇飞. "基于脉搏波的糖尿病和高血压诊断算法研究". 硕士学位论文. 中国矿业大学, 2023.

CASE 10: Corneal edema prediction by PLA 301 Hospital



The predictive accuracy and out-of-bag error rate of each random forest model generated after selecting variables on the basis of copula entropy

医学 3

- 认知医学
 - 认知能力评估 / 痴呆症筛查⁸²
- 运动医学
 - 运动能力评估 / 跌倒风险预测⁸³
 - 重复经颅磁刺激对帕金森病改善神经机制分析⁸⁴
- 精神病学
 - 抑郁症患者识别⁸⁵

⁸² **Jian Ma.** "Predicting MMSE Score from Finger-Tapping Measurement". In: *Proceedings of 2021 Chinese Intelligent Automation Conference*. See also bioRxiv 817338 (2019). 2022. pp. 294–304. ISBN: 978-981-16-6372-7.

⁸³ **Jian Ma**. "Predicting TUG score from gait characteristics based on video analysis and machine learning". In: *Proceedings of 2023 Chinese Intelligent Automation Conference*. See also bioXiv 963686 (2020). 2023, pp. 1–12, **Jian Ma**. "Associations between finger tapping, gait and fall risk with application to fall risk assessment". In: *arXiv preprint arXiv:2006.16648* (2020).

⁸⁴ 李润泽 et al. “重复经颅磁刺激改善帕金森病运动症状的脑功能网络分析”. In: 生物化学与生物物理进展 50.1 (2023), pp. 126–134.

⁸⁵张婷婷 et al. “基于 Couple 熵的抑郁症相干性反馈指标提取”. In: 电子测量技术 45.9 (2022), pp. 160–167.

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药学

- 药物-靶点相互作用预测
 - 提出了一种交互式多特征融合的 DTIs 预测算法⁸⁷

⁸⁷ 高浩田 et al. “基于交互式多特征融合算法的药物靶标预测”. In: 太原理工大学学报 55.4 (2024), pp. 751–758.

- 新冠肺炎流行病 (COVID19)
 - 发热症状疑似病人筛查诊断⁸⁸
- 高血压
 - 高血压关联基因研究⁸⁹

⁸⁹ Soumik Purkayastha and Peter X.K. Song. "Asymmetric predictability in causal discovery: an information theoretic approach". In: *arXiv preprint arXiv:2210.14455* (2022).

● 经济学

- 扶贫政策效果评估, 用于政策目标人口鉴别⁹⁰
- 议价机制中的互惠行为和时间效应⁹¹
- 产业链内部相关性分析⁹²
- 投资者情绪分析⁹³
- 通胀与预期关系分析⁹⁴
- 马来西亚宏观经济分析⁹⁵

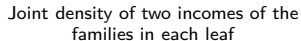
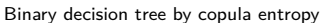
⁹¹ Leonie Bossemeyer. "Machine Learning for Causal Discovery with Applications in Economics". MA thesis. Ludwig-Maximilians-Universität München. 2021.

⁹² 韦颖璐. “基于 pair-copula 熵的相关性度量”. 硕士学位论文. 苏州大学, 2021.

⁹³ Muye Han and Jinsheng Zhou. "Multi-Scale Characteristics of Investor Sentiment Transmission Based on Wavelet, Transfer Entropy and Network Analysis". In: *Entropy* 24.12 (2022), p. 1786. ISSN: 1099-4300. DOI: 10.3390/e24121786. URL: <https://www.mdpi.com/1099-4300/24/12/1786>.

⁹⁴Omid M. Ardakani. "Information Content of Inflation Expectations: A Copula-Based Model". In: *Studies in Nonlinear Dynamics & Econometrics* 29.1 (2024), pp. 71–93. DOI: doi:10.1515/snde-2023-0075. URL: <https://doi.org/10.1515/snde-2023-0075>.

⁹⁵ Nur Layali Mohd Ali Khan and Saiful Izzuan Hussain. "Hubungan Kebersandaran Pasaran Buruh dan Ekonomi Malaysia: Pendekatan Entropi kopula tanpa Andaian Berparameter". In: *Sains Malaysiana* 55.6 (2026), pp. 1067–1076.



- 商品期货价格预测⁹⁶
- 单周期库存管理⁹⁷
- 巴西糖、乙醇和汽油三种大宗商品价格分析⁹⁸
- 中国企业海外并购影响因素分析⁹⁹
- 社会学
 - 分析教育、职业和收入上的性别不平等问题¹⁰⁰

⁹⁷Yu-Xin Tian and Chuan Zhang. "An end-to-end deep learning model for solving data-driven newsvendor problem with accessibility to textual review data". In: *International Journal of Production Economics* 265 (2023), p. 109016. ISSN: 0925-5273. DOI: <https://doi.org/10.1016/j.ijpe.2023.109016>. URL: <https://www.sciencedirect.com/science/article/pii/S09255273232002487>.

⁹⁹ 王琳君,“中国企业海外并购的影响因素和绩效评价研究”,博士学位论文,中国科学院大学,2022。

¹⁰⁰ **Jian Ma.** "Causal Domain Adaptation with Copula Entropy based Conditional Independence Test". In: *arXiv preprint arXiv:2202.13482* (2022). [arXiv: 2202.13482 \[cs.LG\]](https://arxiv.org/abs/2202.13482).

CASE 12: Commodity market analysis by UNESP



Figura 5.0.5: Transfer Entropy entre as séries da volatilidade de preços do açúcar e etanol.



Figura 5.0.6: Transfer Entropy entre as séries da volatilidade de preços do açúcar e gasolina.



Figura 5.0.7: Transfer Entropy entre as séries da volatilidade de preços do etanol e gasolina.

¹⁰⁴ **Bowen Zhang et al.** "Changes in Public Sentiment under the Background of Major Emergencies – Taking the Shanghai Epidemic as an Example". In: *International Journal of Environmental Research and Public Health* 19.19 (2022), p. 12594. ISSN: 1660-4601. DOI: 10.3390/ijerph191912594. URL: <https://www.mdpi.com/1660-4601/19/19/12594>.

CASE 13: LLMs evaluation by Cambridge University

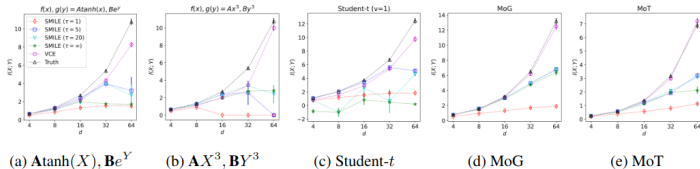


Figure 8: Comparison with the SMILE estimator under different clipping values τ .

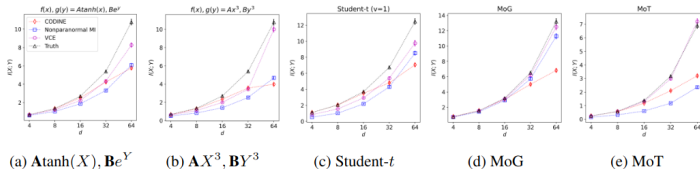


Figure 9: Comparing VCE with classic copula-based estimators e.g. nonparanormal MI (which uses a Gaussian copula to estimate MI) and CODINE (equivalent to classic copula transformation + MINE).

社会科学 4

- 法学
 - 社区属性与社区犯罪关系分析¹⁰⁵
- 政治学
 - 分析政权领导力因素和政权危机之间关系¹⁰⁶
- 军事学
 - 目标意图识别¹⁰⁷
- 情报学
 - 颠覆性技术科学-技术-产业互动模式分析¹⁰⁸
 - 高价值专利评估和发现¹⁰⁹

¹⁰⁵ Mario Wieser. "Learning Invariant Representations for Deep Latent Variable Models". PhD thesis. University of Basel, 2020.

¹⁰⁶ Stuart William Card. "Towards an Information Theoretic Framework for Evolutionary Learning". PhD thesis. Syracuse University, 2011.

¹⁰⁷ 张可 et al. "一种基于动态贝叶斯网络的目标意图识别方法". Pat. CN114997306A. CN114997306A. 2022.

¹⁰⁸ 许海云 et al. "颠覆性技术的科学-技术-产业互动模式识别与分析". In: 情报学报 42.7 (2023), pp. 816–831.

¹⁰⁹ 徐蕊婷, 陈哲宇, and 陈加鹏. "基于两阶段混合式特征选择的潜在高价值专利多情景判别方法研究". In: 数据分析与知识发现 (2026).

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工学 2

● 能源工程 2

- 电网频率稳定性预测¹¹⁵
- 用户线损贡献分析¹¹⁶
- 配电网拓扑辨识¹¹⁷
- 电价预测¹¹⁸
- 电力系统宽频振荡影响因素和传播路径分析¹¹⁹
- 电力系统信息物理融合¹²⁰

¹¹⁵ Peili Liu et al. "Frequency Stability Prediction of Power Systems Using Vision Transformer and Copula Entropy". In: *Entropy* 24.8 (2022), p. 1165. ISSN: 1099-4300. DOI: 10.3390/e24081165. URL: <https://www.mdpi.com/1099-4300/24/8/1165>.

¹¹⁶ Wei Hu et al. "Research on User Loss Contribution Calculation of High-Loss Distribution Area Based on Transfer Entropy". In: *2022 China International Conference on Electricity Distribution (CICED)*. 2022, pp. 499–502. DOI: 10.1109/CICED56215.2022.9929052.

¹¹⁷ 秦超 and 潘毓笙. "一种基于时空特征的配电网拓扑辨识方法". Pat. CN117154679A. CN117154679A. 2023.

¹¹⁸ Xiaoping Xiong and Guohua Qing. "A hybrid day-ahead electricity price forecasting framework based on time series". In: *Energy* 264 (2022), p. 126099. ISSN: 0360-5442. DOI: <https://doi.org/10.1016/j.energy.2022.126099>. URL: <https://www.sciencedirect.com/science/article/pii/S0360544222029851>.

¹¹⁹ 冯双 et al. "一种电力系统宽频振荡影响因素和传播路径分析方法". Pat. CN114977222A. CN114977222A. 2022, 冯双 et al. "基于 Copula 传递熵的设备级和网络级宽频振荡传播路径分析及振荡源定位方法". In: *电工技术学报* 39.16 (2023), pp. 4996–5010. DOI: 10.19595/j.cnki.1000-6753.tces.230873.

¹²⁰ Fengtong Duan and Wanxing Sheng. "Conjugate Information-Energy Entropy Modeling and Entropy Gradient Control Analysis Based on Cyber-Physical Integration". In: *IEEE Transactions on Smart Grid* (2026), pp. 1–1.

CASE 14: Cyber-physical integration by State Grid

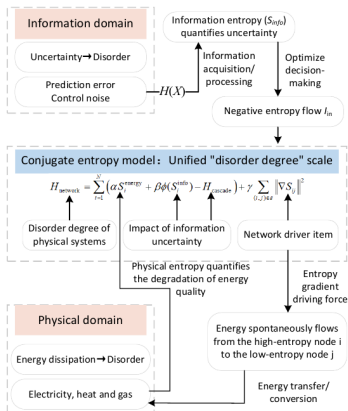


Fig. 1. The core concept diagram of the conjugate entropy theory.

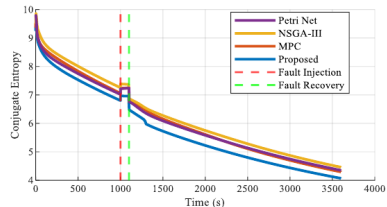


Fig. 7. Comparison of conjugate entropy.

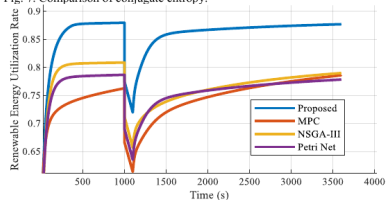


Fig. 8. Comparison of renewable energy utilization rate.

工学 4

- 纺织工程
 - 纱布生产过程质量控制¹²³
 - 可穿戴织物传感器方案设计¹²⁴
- 食品工程
 - 葡萄酒质量与理化成分关系分析¹²⁵

¹²³Seyedeh Azadeh Fallah Mortezaejad et al. "Non-Parametric Multivariate Control Chart Using Copula Entropy". In: *Sankhya B* 87 (2025), pp. 468–518. URL: <https://doi.org/10.1007/s13571-025-00374-y>.

¹²⁴Jiaqi Mo et al. "An Evaluation Framework for Textile Stretch Sensor Layouts Based on Mutual Information". In: *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing*. UbiComp Companion '25. Finland: Association for Computing Machinery, 2026, pp. 35–39. ISBN: 9798400714771. DOI: 10.1145/3714394.3754434. URL: <https://doi.org/10.1145/3714394.3754434>.

¹²⁵Marvin Lasserre, Régis Lebrun, and Pierre-Henri Willemin. "Learning Continuous High-Dimensional Models using Mutual Information and Copula Bayesian Networks". In: *Thirty-Fifth AAAI Conference on Artificial Intelligence, AAAI 2021*. AAAI Press, 2021, pp. 12139–12146. URL: <https://ojs.aaai.org/index.php/AAAI/article/view/17441>, Marvin Lasserre. "Apprentissages dans les réseaux bayésiens à base de copules non-paramétriques". PhD thesis. Sorbonne Université, 2022. URL: <https://tel.archives-ouvertes.fr/tel-03647090>.

CASE 15: Wine quality by Sorbonne Université

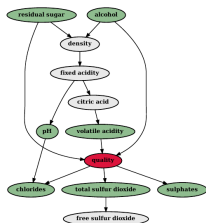


FIGURE 8.10 – Structure reconstruite avec l'algorithme B-CMIIC à partir de l'échantillon des vins rouges. Le paramètre α a été fixé à 0.04 en utilisant la méthode de *cross-validation*. La variable colorée en rouge est la variable cible tandis que l'ensemble des variables colorées en vert correspond à sa couverture de Markov.



FIGURE 8.11 – Structure reconstruite avec l'algorithme B-CMIIC pour l'échantillon des vins blancs. Le paramètre α est fixé à 0.02. La variable colorée en rouge est la variable cible tandis que l'ensemble des variables colorées en vert correspond à sa couverture de Markov.



FIGURE 8.12 – Structure reconstruite avec l'algorithme B-CMIIC pour l'échantillon des vins blancs. Le paramètre α est fixé à 0.06. La variable colorée en rouge est la variable cible tandis que l'ensemble des variables colorées en vert correspond à sa couverture de Markov.

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● 交通运输

- 大件货物运输方案制定¹³¹
- 航空和高铁票价影响因素分析¹³²
- 城市轨道交通客流分析和预测¹³³
- 铁路客流量预测¹³⁴

¹³⁴ Jiahao Chang and Xiaoyu Song. "A Railway Passenger Flow Prediction Model Based on Improved Prophet". In: *Proceedings of the 2023 4th International Conference on Machine Learning and Computer Application*. 2024, pp. 798–804. DOI: 10.1145/3650215.3650354, 常家豪. "基于集成模型的铁路客流量预测研究". 硕士学位论文. 兰州交通大学, 2024.

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- 系统退化过程建模¹⁴¹
- 风电机组健康状态评估¹⁴²
- 砂轮剩余寿命预测¹⁴³
- 滚动轴承剩余使用寿命预测¹⁴⁴

144 Zong Meng et al. "A remaining useful life prediction method of rolling bearings by RSA-BAFT combined with Copula Entropy feature selection". In: *Expert Systems with Applications* 275 (2025). p. 127100.

工学 9

- 石油工程
 - 煤层气井产量预测¹⁴⁵
 - 陆地原油生产碳排放分析¹⁴⁶
- 矿业工程
 - 煤-岩识别¹⁴⁷
 - 重介质选煤¹⁴⁸

¹⁴⁵ **Zhifeng Luo and Haojiang Xi.** "Hybrid Model Based on Copula Mutual Information and SSA-BP: Analysis of Key Factors and Prediction of Stable Gas Production". In: *Arabian Journal for Science and Engineering* 50 (2025), pp. 4673–4685. ISSN: 2191-4281. URL: <https://doi.org/10.1007/s13369-024-09205-0>.

¹⁴⁶Zishang Yuan et al. "An analysis method of influencing factors of crude oil carbon footprint based on XGB-CE: a case study of an onshore oil production area in Shengli Oilfield, China". In: *Clean Technologies and Environmental Policy* 27.11 (2025), pp. 6573–6585. URL: <https://doi.org/10.1007/s10098-025-03203-y>.

¹⁴⁷高峰.“基于多源信息感知的煤-岩识别技术研究”.博士学位论文.武汉大学,2023.

¹⁴⁸建中华 and 代伟. “基于 Copula 熵优化的煤炭重介过程多变量时延参数估计方法”. In: 2024 中国自动化大会. 2024.

工学 10

- 医学工程
 - 脑机接口¹⁴⁹
- 冶金工程
 - 真空蒸馏法制备高纯金属工艺参数优化¹⁵⁰

¹⁴⁹Chao Tang, Dongyao Jiang, and Badong Chen. "MEG Channel Selection Using Copula Entropy-Based Transfer Entropy for Motor Imagery BCI". In: *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. 2024, pp. 1–4. DOI: 10.1109/EMBC53108.2024.10782066, Chao Tang et al. "Copula Transfer Entropy-Based Channel Selection for MEG Motor Imagery Brain Computer Interfaces". In: *Tsinghua Science and Technology* 31.3 (2026), pp. 1474–1486. DOI: 10.26599/TST.2025.9010006. URL: <https://www.sciopen.com/article/10.26599/TST.2025.9010006>.

¹⁵⁰田庆华 et al. "一种真空蒸馏制备高纯金属的优化方法及优化系统". Pat. CN117577229A. CN117577229A. 2024.

工学 11

● 化学工程

- 化学过程故障监测和诊断¹⁵¹
- 化工过程因果网络构建¹⁵²
- 化工过程缺失数据补全¹⁵³
- 生物发酵过程软测量¹⁵⁴

¹⁵¹Min Yin, Jince Li, and Hongguang Li. "A CNN approach based on correlation metrics to chemical process fault classifications with limited labeled data". In: *The Canadian Journal of Chemical Engineering* 101.7 (2022), pp. 3982–3997. doi: 10.1002/cjce.24749. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/cjce.24749>. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/cjce.24749>, Yingpeng Wei and Li Wang. "Copula entropy-based PCA method and application in process monitoring". In: *2022 4th International Conference on Intelligent Information Processing (IIP)*. 2022, pp. 61–64. DOI: 10.1109/IIP57348.2022.00019, Shuangshuang Pan, Li Zhu, and Xirong Xu. "Root cause and fault propagation analysis based on causal graph in chemical processes". In: *2023 CAA Symposium on Fault Detection, Supervision and Safety for Technical Processes (SAFEPROCESS)*. 2023, pp. 1–6. DOI: 10.1109/SAFEPROCESS58597.2023.10295717.

¹⁵²Xiaotian Bi et al. "Large-scale chemical process causal discovery from big data with Transformer-based deep learning". In: *Process Safety and Environmental Protection* 173 (2023), pp. 163–177.

¹⁵³武昊. "基于深度学习的化工过程软测量建模方法研究". 博士学位论文. 北京化工大学, 2023.

¹⁵⁴时和畅. "基于时序差分神经网络的发酵过程批次终点预测及优化研究". 硕士学位论文. 北京化工大学, 2025.

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¹⁶⁸Xu Wang et al. "Physical Layer Secret Key Capacity Using Correlated Wireless Channel Samples". In: *2016 IEEE Global Communications Conference (GLOBECOM)*. 2016. pp. 1–6.

170 Alexandre Delattre, Sylvain Gaeremynck, and Mickaël Corroyer. "Compressing entropy tables with interpolative coding". *Pat.* US12341538B1. US12341538B1. 2024.

¹⁷¹ [Andreas Gocht-Zech](#). "Ein Framework zur Optimierung der Energieeffizienz von HPC-Anwendungen auf der Basis von Machine-Learning-Methoden". [PhD thesis](#). Technische Universität Dresden, 2022.

¹⁷²Lin Liu, Cong Hu, and Xiao-Jun Wu. “ CE^2 : A Copula Entropic Mutual Information Estimator for Enhancing Adversarial Robustness”. In: *Pattern Recognition and Computer Vision*. Springer Singapore, 2024, pp. 163–174.

174 曹久慧 et al. “一种基于自注意力机制的变形监测方法”. Pat. CN116378120A. CN116378120A. 2023, 张旭辉. “顾及邻近点相似性的机器学习变形分析方法研究”. 硕士学位论文. 武汉大学, 2022.

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金融工程 1

- 投资组合优化
 - 股票资产相关性网络分析¹⁷⁶
 - ST 股票分类¹⁷⁷
- 金融问题建模
 - Copula 函数模型选择¹⁷⁸
- 股票相关性建模
 - R-vine copula 结构建模¹⁷⁹

¹⁷⁶ Qitong Wang. "Social Networks, Asset Allocation and Portfolio Diversification". MA thesis. University of Waterloo, 2015, 廖轶楠. "基于 Copula 熵选股及集成神经网络预测的投资组合管理研究". 硕士学位论文. 南京信息工程大学, 2023.

¹⁷⁷ 朱仲儿. "多种机器学习方法的股票分类预测". 硕士学位论文. 上海师范大学, 2022.

¹⁷⁸ Rafael Calsaverini and Renato Vicente. "An information-theoretic approach to statistical dependence: Copula information". In: *EPL (Europhysics Letters)* 88.6 (2009), p. 68003. DOI: 10.1209/0295-5075/88/68003. URL: <https://doi.org/10.1209/0295-5075/88/68003>.

¹⁷⁹ Fadhah Amer Alanazi. "Truncating Regular Vine Copula Based on Mutual Information: An Efficient Parsimonious Model for High-Dimensional Data". In: *Mathematical Problems in Engineering* 2021 (2021), p. 4347957. URL: <https://doi.org/10.1155/2021/4347957>, 王念鸽. "基于互信息的 Vine Copula 模型的高频数据投资组合风险测度研究". 硕士学位论文. 浙江财经大学, 2023.

金融工程 2

- 量化金融工具箱 MLFinLab¹⁸⁰
- 金融系统性风险
 - 行业风险溢出效应分析¹⁸¹
 - 金融脆弱性度量¹⁸²
- 信用风险评价
 - 信用风险卡模型建立¹⁸³
 - P2P 借贷个人信用风险模型¹⁸⁴

¹⁸⁰ Hudson and Thames. *Machine Learning Financial Laboratory (MLFinLab)*. GitHub. 2021. URL: <https://github.com/hudson-and-thames/mlfinlab>.

¹⁸¹ 熊靖宇. “基于 Copula 熵的行业风险溢出效应分析”. 硕士学位论文. 东北财经大学, 2020.

¹⁸² Wanfang Chen and Marc G Genton. “Are you all normal? It depends!” In: *International Statistical Review* 91.1 (2023), pp. 114–139.

¹⁸³ 孔祥永 et al. “一种自动化特征工程信用风险评价系统及方法”. Pat. CN114049198A. CN114049198A. 2021.

¹⁸⁴ 彭翊庭. “个人信用风险评估模型比较——基于 Copula 熵的特征选择”. 硕士学位论文. 清华大学, 2022.

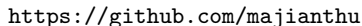
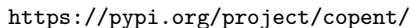
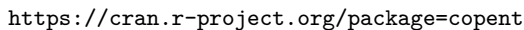
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<https://doi.org/10.1007/s00521-023-09047-1>.

COVID-19 pandemic". In: *Statistics in Transition new series* 25.1 (2024), pp. 23–41. DOI: 10.59170/stattrans-2024-002.

The **copent** package in R and Python for estimating copula entropy, transfer entropy, the statistic for multivariate normality test, two-sample test, and change point detection are available on CRAN and PyPI respectively. The source codes are provided on GitHub.



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Softwares

Third-party

- R: `cylcop`¹⁸⁹ (by University of Zürich and University of Minnesota);
- Python: `MLFinLab`¹⁹⁰ and `ArbitrageLab`¹⁹¹ (authored by Prof. Marcos López de Prado from Cornell University, Github 4500+ and 670+ stars);

¹⁸⁹Florian H. Hodel and John R. Fieberg. "cylcop: An R Package for Circular-Linear Copulae with Angular Symmetry". In: *bioRxiv* (2021), p. 2021.07.14.452253, Florian Hodel. *cylcop: Circular-Linear Copulas with Angular Symmetry for Movement Data*. CRAN. R package version 0.2.0, URL: <https://cran.r-project.org/package=cylcop>. 2022.

¹⁹⁰Hudson and Thames. *Machine Learning Financial Laboratory (MLFinLab)*. GitHub. 2021. URL: <https://github.com/hudson-and-thames/mlfinlab>.

¹⁹¹Hudson and Thames. *The ArbitrageLab package in Python*. GitHub. 2024. URL: <https://github.com/hudson-and-thames/mlfinlab>.

Softwares

Third-party

- Python: **gcmi**¹⁹² (by Glasgow University, Radboud University Nijmegen, University of Amsterdam, Kyiv AI group, Github 100+ stars), **pytorch-mighty**¹⁹³ (by Kyiv AI group), **HOI**¹⁹⁴ and **Frites**¹⁹⁵ (by CNRS, INT, and Aix-Marseille Université), **Tensorpac**¹⁹⁶ (by Université de Montréal, INT, and Aix-Marseille Université), **driada**¹⁹⁷ (by Moscow State University);

¹⁹²Robin A.A. Ince et al. "A statistical framework for neuroimaging data analysis based on mutual information estimated via a gaussian copula". In: *Human Brain Mapping* 38.3 (2017), pp. 1541–1573, Robin A.A. Ince et al. *gcmi* : *Gaussian-Copula Mutual Information*. GitHub. URL: <https://github.com/robinince/gcmi>. 2020.

¹⁹³Danylo Ulianych. *The pytorch-mighty package in Python*. GitHub. 2023. URL: <https://github.com/dizcza/pytorch-mighty>.

¹⁹⁴BraiNets. *The HOI package in Python*. Github. 2024. URL: <https://brainets.github.io/hoi/>.

¹⁹⁵Etienne Combrisson et al. "Frites: A Python package for functional connectivity analysis and group-level statistics of neurophysiological data". In: *Journal of Open Source Software* 7.79 (2022), p. 3842. DOI: 10.21105/joss.03842. URL: <https://doi.org/10.21105/joss.03842>.

¹⁹⁶Etienne Combrisson et al. "Tensorpac: An open-source Python toolbox for tensor-based phase-amplitude coupling measurement in electrophysiological brain signals". eng. In: *PLoS computational biology* 16.10 (2020), e1008302. ISSN: 1553-7358. DOI: 10.1371/journal.pcbi.1008302.

¹⁹⁷Institute for Advanced Brain Studies. *driada: Dimensionality Reduction for Integrated Activity Data*. Github. URL: <https://github.com/iabs-neuro/driada>. 2024.

Softwares

Third-party

- Python: CopulaGP¹⁹⁸ (by University of Edinburgh), **Polars-ds**¹⁹⁹ (by New York University, 1.45+ million downloads, Github 650+ stars), **copul**²⁰⁰ (by Albert-Ludwigs-Universität Freiburg), **rcopula**²⁰¹, and **pycopcor**²⁰² (by Technische Universität Dresden);

¹⁹⁸Nina Kudryashova et al. "Parametric Copula-GP model for analyzing multidimensional neuronal and behavioral relationships". In: *PLoS computational biology* 18.1 (2022), e1009799, Nina Kudryashova. *Parametric Copula-GP framework*. Github. URL: <https://github.com/NinelK/CopulaGP>. 2022.

¹⁹⁹Tianren Qin. *Polars-ds: Polars Extension for General Data Science Use*. Github. URL: https://github.com/abstractqqq/polars_ds_extension. 2024.

²⁰⁰Marcus Rockel. *The copul package in Python*. Github. URL: <https://github.com/Corrram/copul>. 2026.

²⁰¹Christopher Bell. *The rcopula package in Python*. Github. URL: <https://github.com/Chrisebell124/python-copula>. 2026.

²⁰²Andreas Gocht-Zech. *The pycopcor package in Python*. Github. URL: <https://github.com/AndreasGocht/pycopcor>. 2023.

- Julia: GenEnt_11²⁰³ (the University of Edinburgh) ConcolityTools_11²⁰⁴

²⁰⁹The CE estimation method is decomposed into two functions for empirical copula estimation and entropy estimation.

Thanks!

Q&A